Predictive Early Warning System for Rice Productivity

Addressing Indonesia food security risks and farmer income instability caused by climate change

Mutiara Hernowo | 35236043

Problem & Objectives

Problem

Significant changes in rainfall and temperature directly impacts rice productivity.

Predict

Predict provincial annual productivity using ML modelling and deliver an Early Warning System (EWS)

Warn

2–3 months before harvest to enable proactive interventions

Act

Phenology by proxy + Leave-One-Year-Out (LOYO) temporal validation

Research Gap

Rice production and climate relationships are non-linear with threshold effects (e.g., heat during flowering)

Prior Work

Older models were often too linear, overly complex, or difficult to deploy locally, failing to capture critical non-linear climate impacts.

Our Solution

Phenology by proxy using annual climate indicator aggregations.
This keeps the model lightweight and deployable by local governments and agricultural extension workers.

We compare multiple techniques: **OLS, Random Forest, and XGBoost** to find the optimal balance of accuracy and efficiency.

Dataset Overview

10 Provinces with High Rice Production Rate (Java, Sumatera, Sulawesi)

A tibble: 23,113 × 17

date <date>	tn <dbl>	tx <dbl>	tavg <dbl>	rh_avg <dbl>	rr <dbl>	ss <dbl>	ff_x <dbl>	ddd_x <dbl>	ff_avg <dbl>
2019-01-01	21.2	33.80000	29.00000	70.00000	0.00	7.4	4	120	2
2019-01-01	22.2	30.00000	25.40000	88.00000	16.10	2.1	4	250	3
2019-01-01	24.8	31.00000	27.80000	82.00000	3.20	5.4	8	350	3
2019-01-01	21.8	28.80000	24.60000	82.00000	0.00	7.2	4	50	1
2019-01-01	25.4	33.38328	26.80000	87.00000	3.50	6.4	1	200	0
2019-01-01	23.8	31.60000	28.30000	78.00000	0.00	6.0	6	45	1
2019-01-01	23.6	30.20000	25.80000	88.00000	11.40	3.9	7	220	2
2019-01-01	24.6	32.60000	28.00000	86.00000	0.20	0.1	4	310	2
2019-01-01	22.5	30.40000	25.00000	89.00000	9.30	1.9	10	286	2
2019-01-01	21.6	31.00000	25.10000	87.00000	1.80	0.9	3	80	1

1–10 of 23,113 rows | 1–10 of 17 columns   Previous 1 2 3 4 5 6 ... 100 Next

BMKG Climate Dataset

Source: https://dataonline.bmkg.go.id/

Volume: 23,113 observations and 17 variables then aggregated by province and year → 60 obs. and 8 variables (annual data)

Features:

- Year
- Province
- Temperature (max, avg)
- Humidity rate
- Rainfall
- Hot days frequency
- Sunshine duration

A tibble: 60 × 5

year <dbl>	province <chr>	yield_area <dbl>	field_prod <dbl>	rice_prod <dbl>
2019	Aceh	310012.5	55.30	1714438
2019	Banten	303731.8	48.41	1470503
2019	Central Java	1678479.2	57.53	9655654
2019	East Java	1702426.4	56.28	9580934
2019	Lampung	464103.4	46.63	2164089
2019	North Sumatera	413141.2	50.32	2078902
2019	South Sulawesi	1010188.8	50.03	5054167
2019	South Sumatera	539316.5	48.27	2603396
2019	West Java	1578835.7	57.54	9084957
2019	West Sumatera	311671.2	47.58	1482996

1–10 of 60 rows   Previous 1 2 3 4 5 6 Next

BPS Rice Production Dataset

Source: https://www.bps.go.id/

Volume: 60 observations and 5 variables

Features:

- Year
- Province
- Yield area (ha) → 1 ha = 10,000 sqm
- Yield productivity (qu/ha)
- Rice production (ton)

A tibble: 60 × 15

year <dbl>	province <chr>	yield_area <dbl>	field_prod <dbl>	rice_prod <dbl>	rain_sum <dbl>	temp_avg <dbl>	hot_days <int>	humid_avg <dbl>
2019	Aceh	310012.5	55.30	1714438	1490.55	29.00055	94	73.24242
2019	Banten	303731.8	48.41	1470503	1972.90	28.25192	75	76.76923
2019	Central Java	1678479.2	57.53	9655654	1249.40	28.50472	122	76.36798
2019	East Java	1702426.4	56.28	9580934	1884.80	23.98767	0	76.23014
2019	Lampung	464103.4	46.63	2164089	2431.00	27.26960	96	79.66193
2019	North Sumatera	413141.2	50.32	2078902	1895.70	28.88791	50	77.28022
2019	South Sulawesi	1010188.8	50.03	5054167	1858.10	27.57983	35	77.28453
2019	South Sumatera	539316.5	48.27	2603396	2047.30	27.84356	114	85.89315
2019	West Java	1578835.7	57.54	9084957	5276.40	26.30674	25	81.00281
2019	West Sumatera	311671.2	47.58	1482996	5307.70	25.62486	5	87.72099

1–10 of 60 rows | 1–9 of 15 columns   Previous 1 2 3 4 5 6 Next

Combined Dataset

Combined both dataset (Year and Province)

Volume: 60 observations and 15 variables

Features:

- Year
- Province
- Yield area (ha) → 1 ha = 10,000 sqm
- Yield productivity (qu/ha)
- Rice production (ton)
- Rainfall rate
- Temperature average
- Hot days frequency
- Humidity rate
- Sunshine
- Hot/wet and rain/dry season
- Flowering days

Feature Engineering (Proxy)

F0

rain_sum, temp_avg, humid_avg, sunshine_sum, hot_days

→

F1

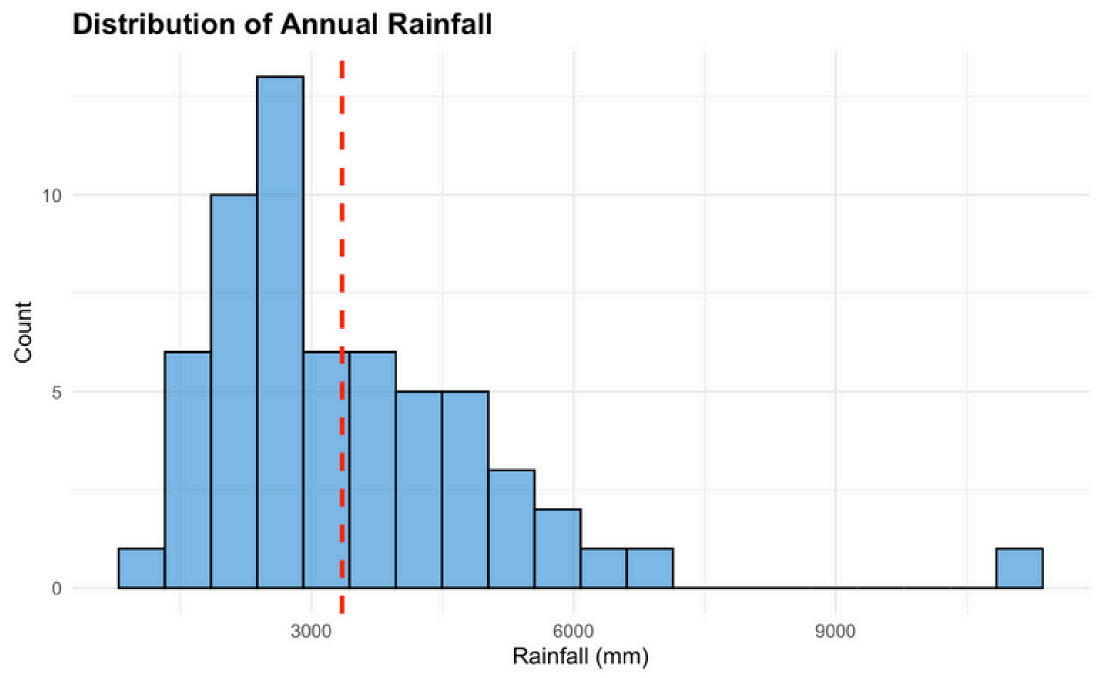
F0 +
wet/dry season aggregates

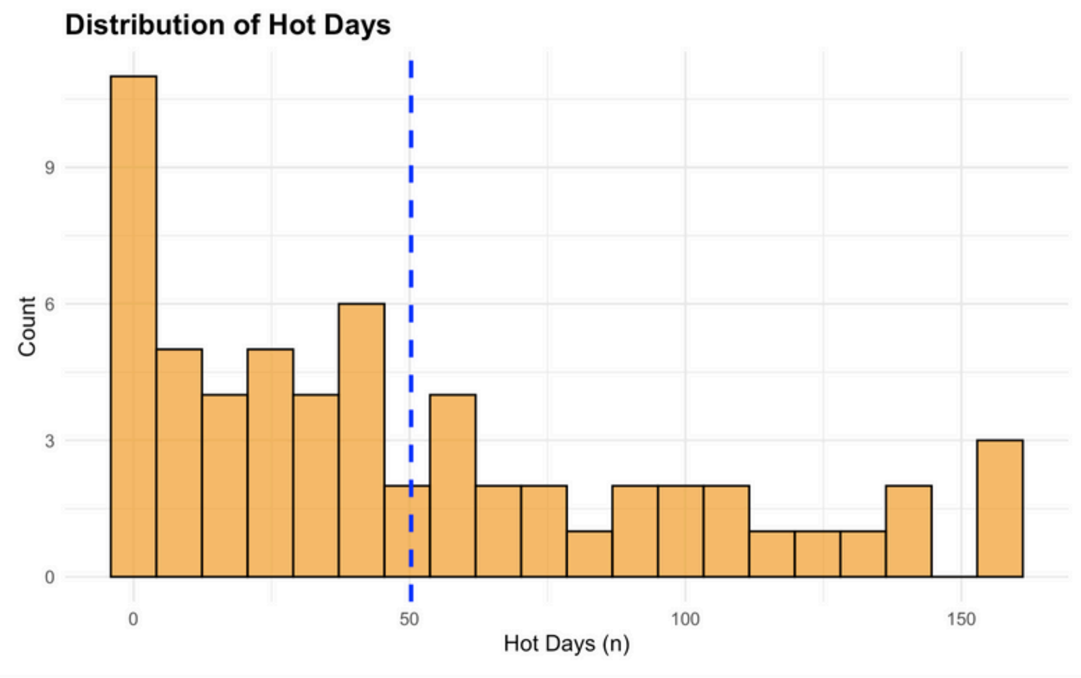
→

F2

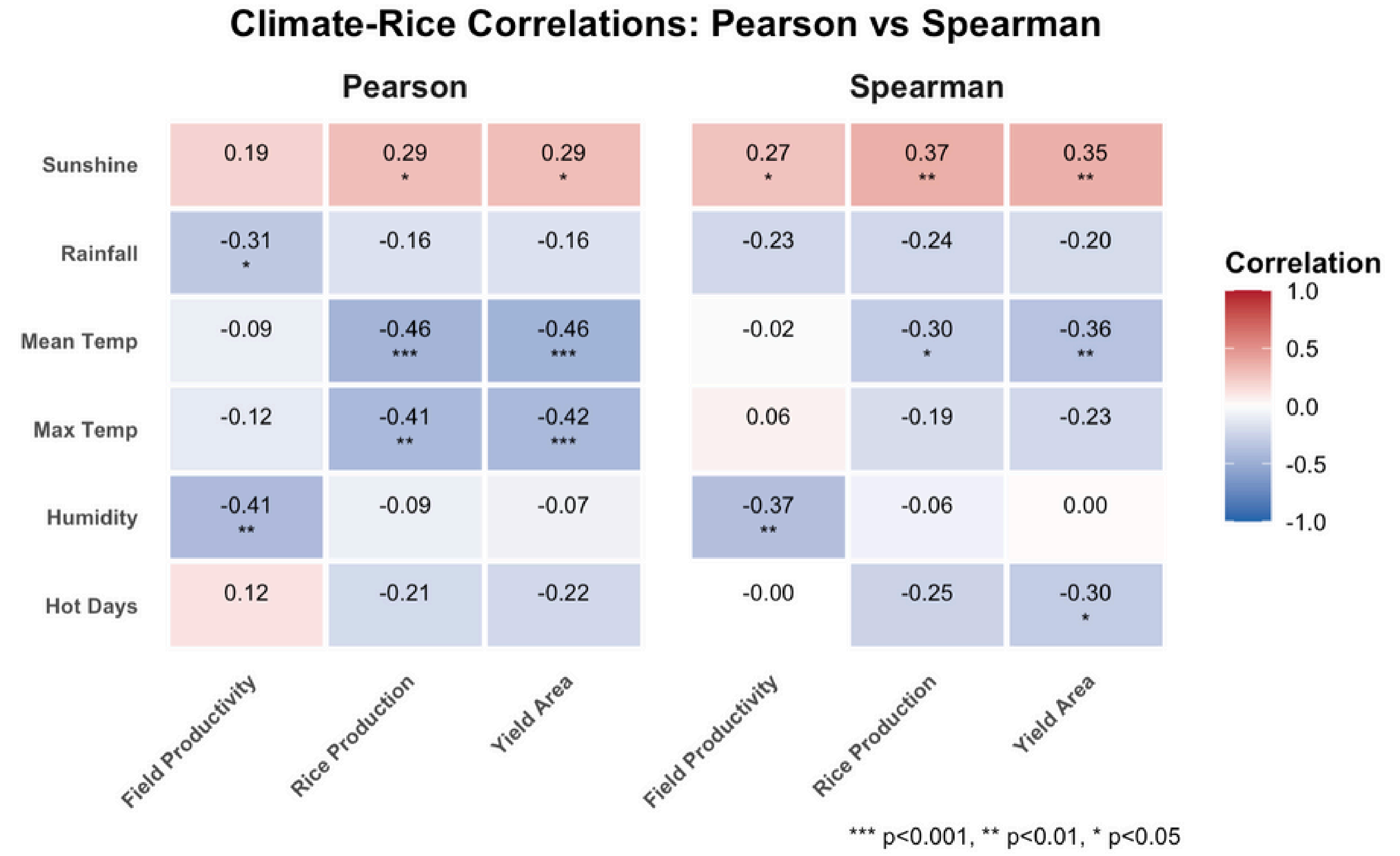
F1 +
flowering-window heat
(Jan–Mar & Jun–Aug)

Exploratory Data Analysis





Rainfall is right-skewed, while hot_days is long-tailed and varies by province



Spearman shows **stronger monotonic** links than Pearson; hot_days ↘ productivity Signals support non-linear models and seasonal features.

Modelling and Evaluation

Algorithm Tested

Ordinary Least Square (OLS)
Random Forest (RF)
eXtreme Gradient Boosting (XGBoost)

Validation

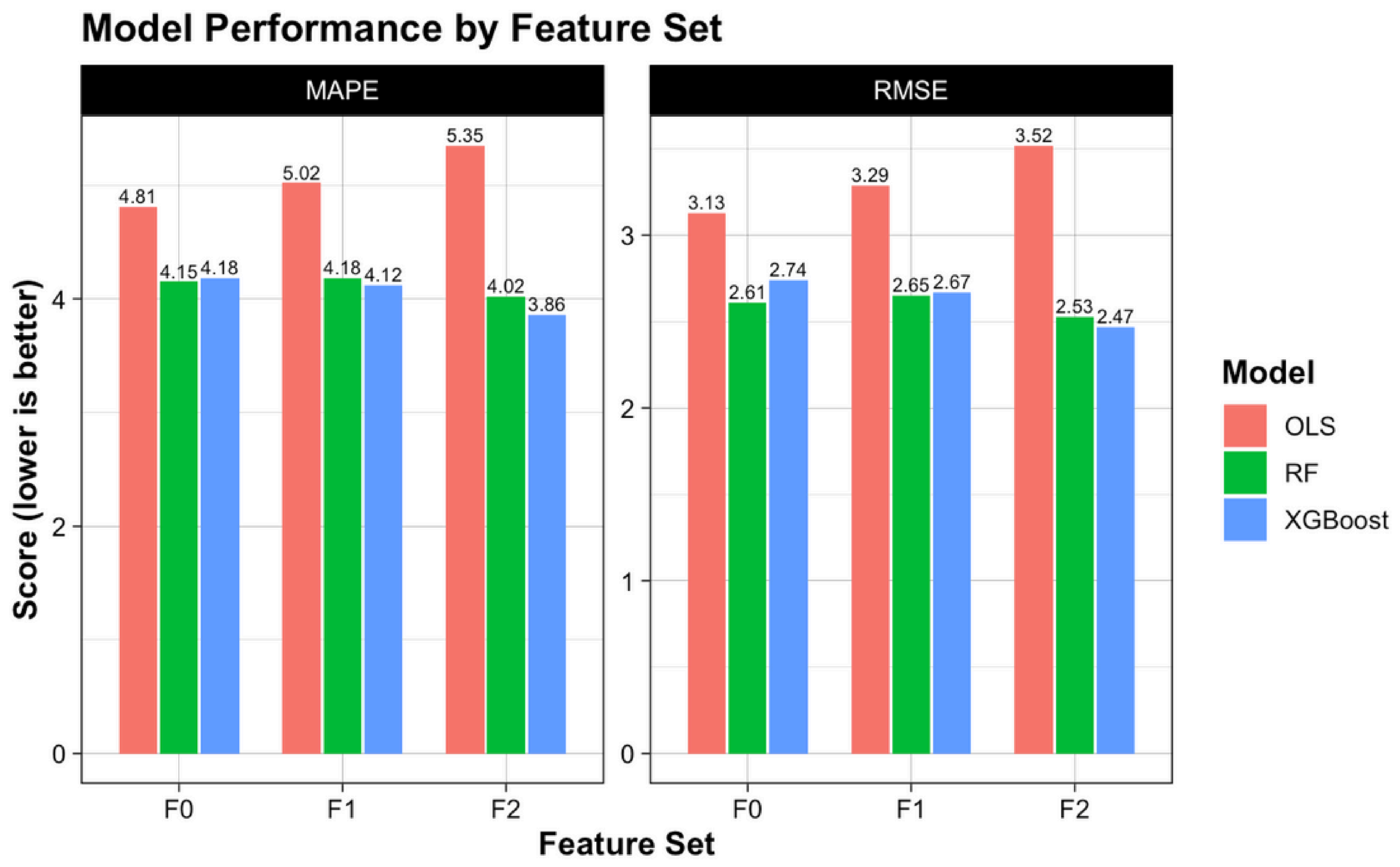
Leave-One-Year-Out (LOYO) cross-validation:
Training on 5 years and testing on the remaining 1 year, averaged across all possible temporal splits

Evaluation

Root Mean Square Error (RMSE in ton/ha)

Mean Absolute Percentage Error (MAPE in %).

Result

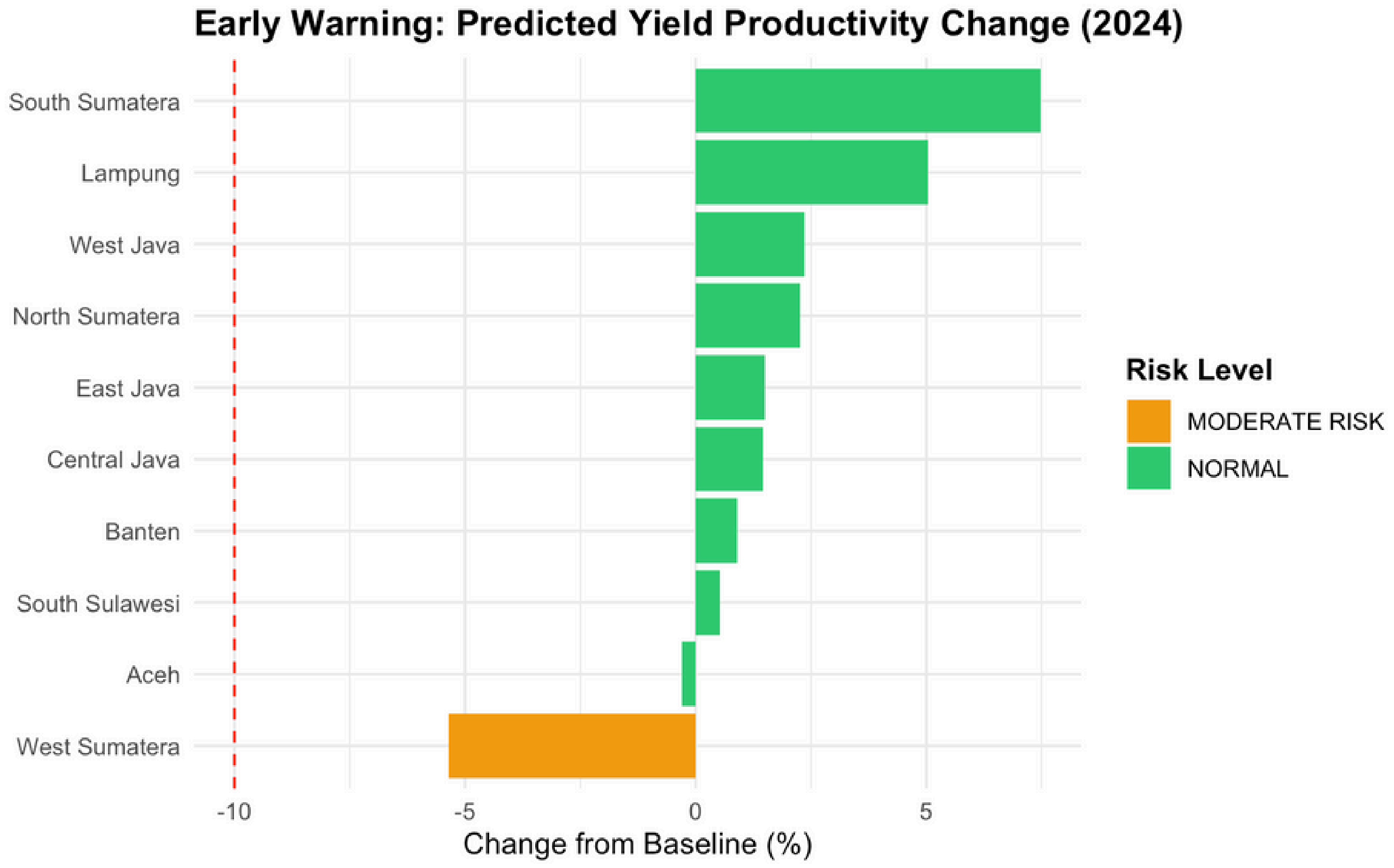


Best model: XGBoost + F2 → RMSE 2.38, MAPE 3.71%.
vs F0: RMSE ↓ ~11%, MAPE ↓ ~14% (temporal generalisation improves).
Influential features: hot_days, temp_max, rainfall.

Early Warning System

Predict 2024 productivity; compare to multi-year baseline per province

province <chr>	predicted_prod <dbl>	baseline_prod <dbl>	change_in_productivity <dbl>	risk <chr>
West Sumatera	45.93979	48.540	−5.3568422	MODERATE RISK
Aceh	55.11027	55.276	−0.2998201	NORMAL
South Sulawesi	50.64970	50.392	0.5113986	NORMAL
Banten	51.84052	51.380	0.8963000	NORMAL
Central Java	57.18997	56.372	1.4510184	NORMAL
East Java	57.32964	56.486	1.4935305	NORMAL
North Sumatera	52.56017	51.398	2.2611102	NORMAL
West Java	58.46980	57.126	2.3523489	NORMAL
Lampung	52.49989	49.984	5.0333894	NORMAL
South Sumatera	55.82974	51.942	7.4847762	NORMAL



Risk tiers:

change ≤ −5% ⇒ MODERATE
otherwise NORMAL (tunable)

Standards & Governance

Process

CRISP-DM methodology ensures a structured approach. All outputs are generated via a reproducible R Markdown pipeline with full version control

Data Governance

Includes automated schema and range checks, documentation of the mean imputation strategy, and reliance on public data, verifiable sources (BMKG/BPS)

Model Governance

LOYO reports, RMSE/MAPE tracking, F0–F2 ablation, risk-threshold documentation & change control

Future Development

DBMS to carry large dataset
38 provinces

Real-time EWS
Apache Kafka